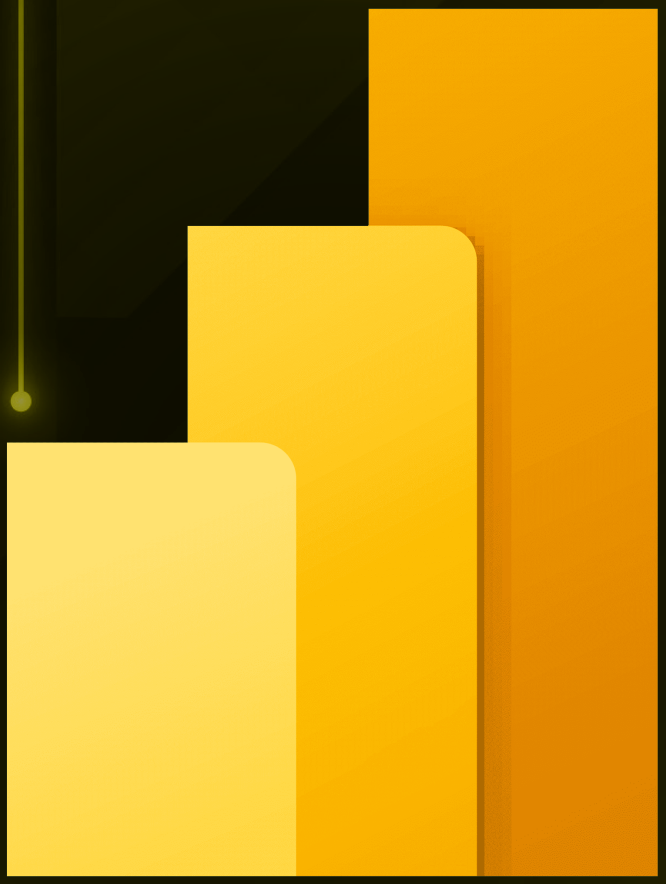


POWER BI

Lecture: Sum Chanthou

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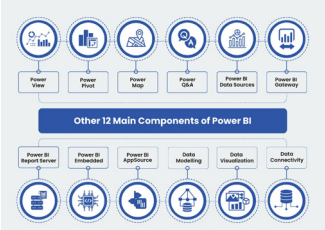
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Chapter 1: Introduction to Power BI



Power BI is Microsoft's business intelligence platform that allows organizations and individuals to connect to data, transform it into meaningful insights, and share those insights across teams. It provides an end-to-end solution for analyzing data, creating interactive reports, and making data-driven decisions.

1.1. Key components



Power BI is made up of several important components that work together to deliver a complete analytics solution. Each plays a unique role in the process of transforming raw data into actionable insights.

1.1.1. Power BI Desktop

- A **free Windows application** for building reports and data models.
- **Users** can connect to data sources, clean and transform data using Power Query, and create dashboards with visualizations.
- This is often where most of the report design work is done before publishing to the cloud.

1.1.2. Power BI Service

- A cloud-based platform where reports and dashboards can be shared, viewed, and collaborated on.
- Provides features like workspaces, apps, and data refresh scheduling.
- Makes insights accessible to users through web browsers or mobile apps.

1.1.3. Power BI Mobile

- A mobile application (iOS, Android, Windows) for accessing and interacting with Power BI reports on the go.
- Ensures decision-makers can view up-to-date data anytime, anywhere.

1.2. Advantages

Power BI has become one of the most popular business intelligence tools because it combines ease of use with powerful data analysis capabilities. Below are some of its key advantages.



1.2.1. Integrates data from multiple sources

- **Power BI** connects to a wide variety of data sources, including Excel spreadsheets, SQL Server databases, cloud services like Azure and Salesforce, and even web-based APIs.

1.2.2. Easy-to-use visuals for better decision-making

- A mobile application (iOS, Android, Windows) for accessing and interacting with Power BI reports on the go.
- Ensures decision-makers can view up-to-date data anytime, anywhere.

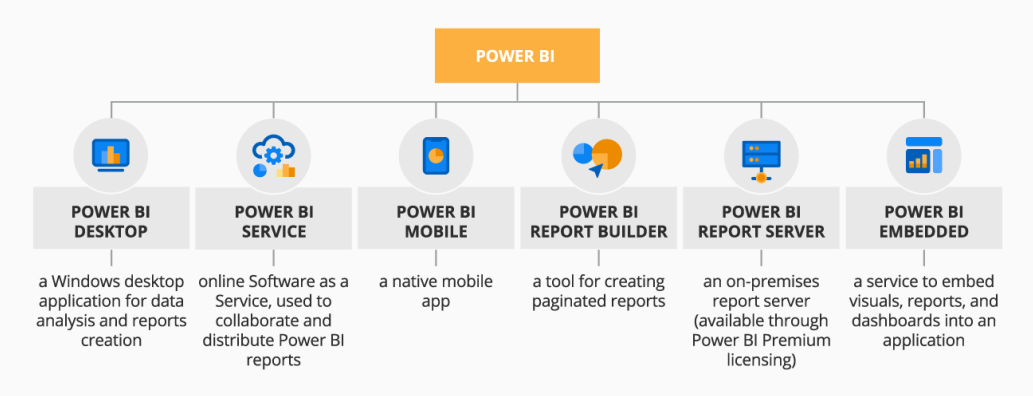
1.2.3. Real-time data updates

- With Power BI, dashboards can be set up to refresh automatically, ensuring that users always have the most up-to-date information.Ensures decision-makers can view up-to-date data anytime, anywhere.
- This is especially valuable for monitoring live business operations, such as sales, inventory, or financial performance.

1.2.4. Supports collaboration across teams

- Power BI Service enables reports and dashboards to be shared easily within an organization.
- Teams can collaborate by working together in shared workspaces, adding comments, and using version control features.

1.3. Core concepts



To fully understand how Power BI works, it is important to learn its fundamental building blocks. These core concepts—datasets, reports, dashboards, and tiles—form the foundation of the Power BI environment.

1.3.1. Dataset

- A dataset is a collection of data imported or connected to Power BI from various sources (Excel, SQL Server, cloud services, etc.).
- Datasets can be transformed and cleaned using Power Query before being used in reports.

1.3.2. Report

- A report is an interactive, multi-page collection of visuals created from a dataset.
- Reports allow users to drill down, filter, and explore data in detail.

1.3.3. Dashboard: A single-page view combining multiple visuals

- A dashboard is a high-level, single-page view that brings together key metrics and visuals from one or more reports or datasets.
- Dashboards are often designed for quick monitoring, giving managers a snapshot of business performance.

1.3.4. Tiles: Individual visualizations on dashboards

- A tile is a single visualization (such as a chart, graph, or KPI indicator) that appears on a dashboard.
- Tiles can be pinned from reports or created directly in dashboards.

Chapter 2: Getting Started with Power BI Desktop

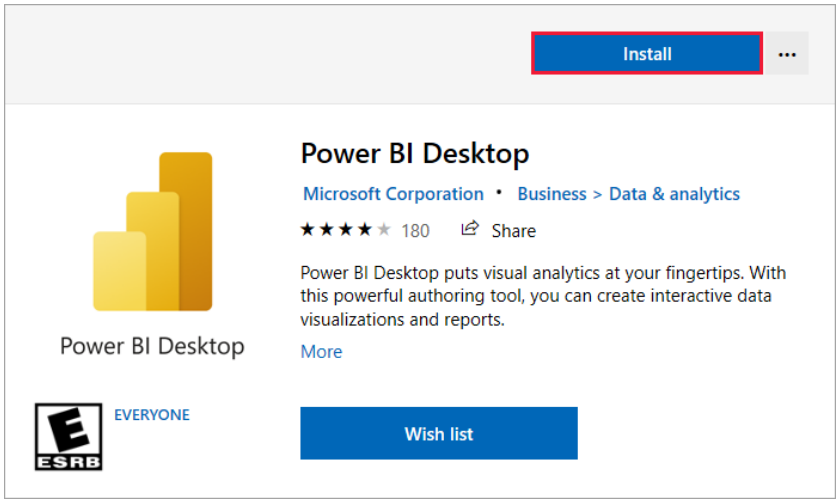


Power BI Desktop is the primary tool for creating reports and visualizations in Power BI. It is a free Windows application that allows users to connect to data sources, transform data, and build interactive reports before publishing them to the Power BI Service.

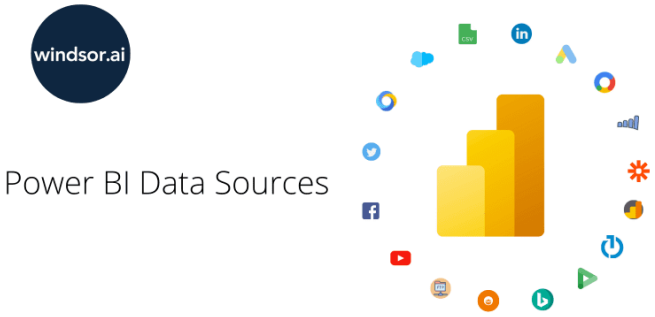
This chapter introduces the basic workflow of Power BI Desktop and explains how to begin your journey with the tool.

2.1. Installation and Launch

- Power BI Desktop can be downloaded for free from the Microsoft Store or the official Power BI website.
- Installation is straightforward and similar to other Windows applications.



2.2. Connecting to Data Sources



When you open Power BI Desktop, you will find several key areas:

- Ribbon – Contains commands for data connections, modeling, and visualizations.
- Report View – The main canvas where charts and visuals are placed.
- Data View – Displays the tables and data you've loaded.
- Model View – Shows relationships between tables in your dataset.

2.3. Power Query Editor

Power BI Desktop supports a wide range of data sources, including:

- Files (Excel, CSV, XML, JSON)
- Databases (SQL Server, Oracle, MySQL, etc.)
- Cloud Services (Azure, Google Analytics, Salesforce)
- Online Services (SharePoint, Dynamics 365, etc.)

The Get Data option on the ribbon is used to connect to these sources.
Once connected, data can be previewed before loading.



2.4. Creating Visualizations



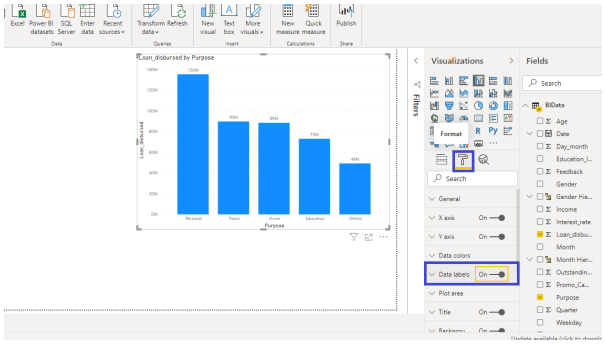
After connecting to data, Power BI allows cleaning and transformation using the **Power Query Editor.**

Common transformations include:

- Removing duplicates or blank rows
- Filtering data
- Merging tables
- Splitting or combining columns
- Changing data types

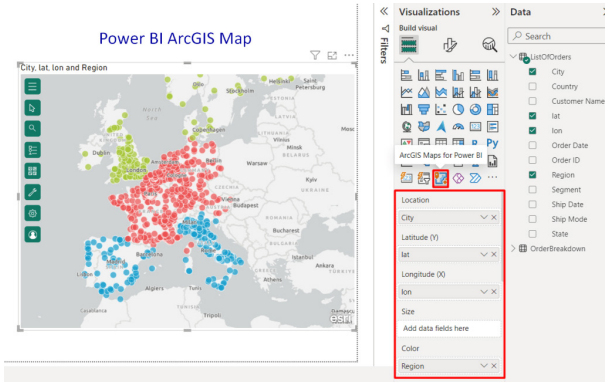
Transformations are recorded as steps and can be modified later without affecting the original data.

2.4.1. Charts: Bar, Column, Line, Pie



- Commonly used for sales by product, revenue by region, or employee performance comparisons.
- Best suited for showing trends over time, such as monthly revenue, website traffic, or temperature changes.
- Donut charts provide a variation with a hole in the middle, which can be used to display additional information.

2.4.2. Maps: Geographical representation of data



- Power BI includes map visuals that display data geographically.
- By using location fields such as country, state, or postal code, data points can be plotted on a map.
- Types of map visuals:
  - Basic Map – plots bubbles on a map based on data values.
  - Filled Map (Choropleth) – colors regions (countries, states, etc.) based on values.

2.4.3. Tables and matrices

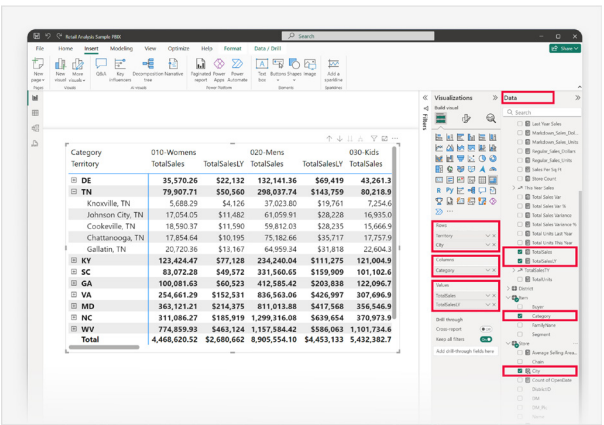


Table Visuals

- Display raw data in rows and columns, similar to an Excel spreadsheet.
- Useful for showing detailed records, such as transactions or invoices.

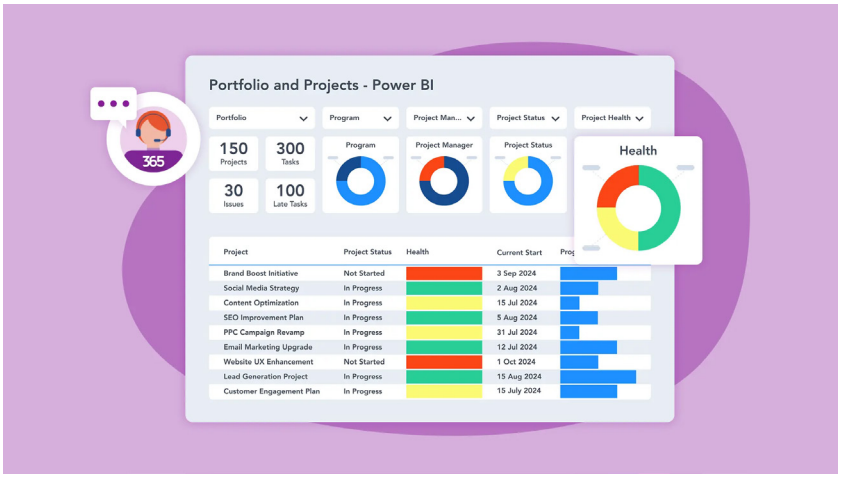
Matrix Visuals

- Display raw data in rows and columns, similar to an Excel spreadsheet.
- Useful for showing detailed records, such as transactions or invoices.

Useful for financial statements, sales performance across time, or multi-level data comparisons.

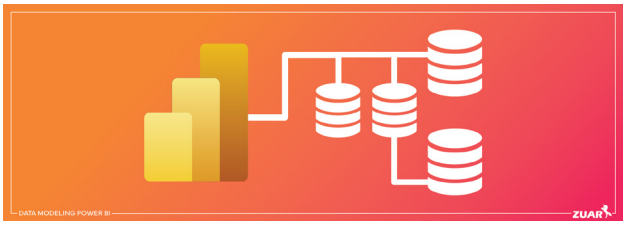
2.5. Saving and Publishing Reports

- Once data is ready, users can drag fields onto the canvas to create visualizations.
- Visuals can be customized with formatting options such as colors, labels, and filters.
- Multiple report pages can be created to organize insights by topic (e.g., Sales, Finance, Operations).
- Reports created in Power BI Desktop are saved with a .pbix file extension.



These files can be published directly to the Power BI Service, where they can be shared with others. Publishing ensures collaboration and access to reports via the web and mobile apps.

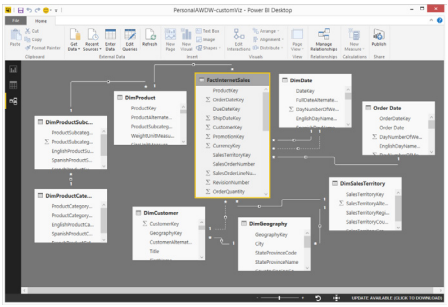
Chapter 3: Data Modeling and DAX





To create meaningful insights, Power BI requires more than just importing data. Effective analysis depends on proper data modeling and the use of DAX (Data Analysis Expressions) to build calculated fields, measures, and logic. This chapter introduces the fundamentals of data modeling and DAX to help build strong analytical foundations.

3.1. Data Modeling



Organize Datasets and Define Relationships

Data modeling is the process of structuring datasets in a way that supports efficient analysis. In Power BI, this involves creating relationships between tables, defining hierarchies, and choosing the right schema design.

3.1.1. One-to-many and many-to-one relationships

- In most cases, Power BI relationships are one-to-many (1:)* or many-to-one (*:1).
- Example: A single customer (one) may have multiple orders (many).
- Relationships allow Power BI to connect tables and aggregate data correctly across different sources.

3.1.2. Use star or snowflake schema for efficiency

Star Schema

- Consists of a central fact table (e.g., Sales) connected to multiple dimension tables (e.g., Customers, Products, Dates).
- Simpler, faster, and most commonly recommended for Power BI.

Snowflake Schema

- A variation where dimension tables are normalized into multiple related tables.
- More complex but can reduce data duplication.

Power BI generally performs best with a star schema because it is easier to maintain and query.

3.2. DAX (Data Analysis Expressions)



DAX is the formula language used in Power BI to perform calculations on data models. It enables users to create measures, calculated columns, and advanced expressions for analysis.

3.2.1. Measures:

Calculated values evaluated dynamically

- Measures are dynamic calculations that are evaluated based on filters in a report.
- Example:

  Total Sales = SUM(Sales[Amount])
- Measures are ideal for aggregations such as totals, averages, and ratios.

3.2.2. Calculated Columns

- A calculated column is added directly to a table and computed row by row.
- Example:

  Profit = Sales[Revenue] - Sales[Cost]
- Unlike measures, calculated columns increase dataset size, so they should be used only when necessary.

3.2.3. Common functions

3.2.3.1. **SUM, AVERAGE, COUNT**

- SUM(Sales[Amount]) → Adds up all values.
- AVERAGE(Sales[Amount]) → Finds the mean.
- COUNT(Customer[ID]) → Counts rows or distinct values.

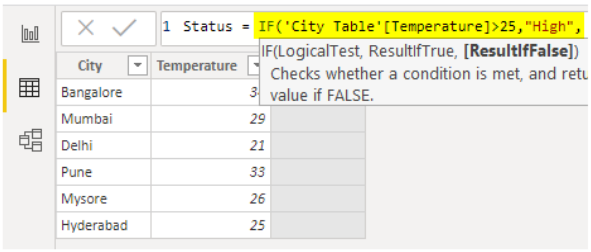
3.2.3.2. **IF** statements for conditional logic

- IF(Sales[Amount] > 1000, “High”, “Low”)
- Allows conditional outputs based on logical tests.

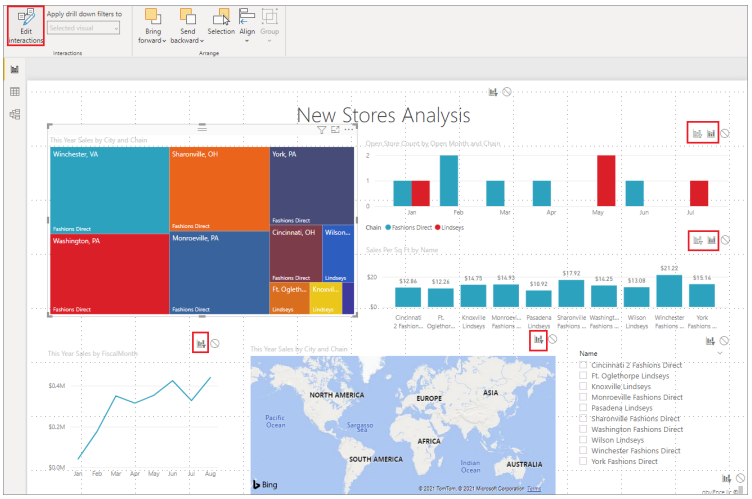
3.2.3.3. CALCULATE for advanced calculations

- CALCULATE modifies the filter context to create more complex measures.
- Example:

  SalesLastYear = CALCULATE(SUM(Sales[Amount]), YEAR(Sales[Date]) = 2024)
- This makes it one of the most powerful DAX functions for time intelligence and scenario-based calculations.



3.3. Interactive Visuals

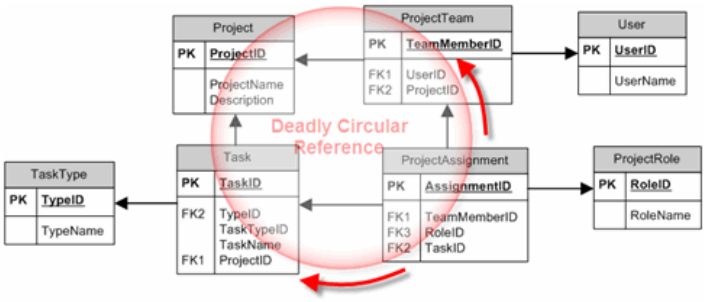


With a solid data model and DAX, Power BI reports become more interactive.

Users can slice and filter data dynamically, drilling down into hierarchies (e.g., Year → Month → Day).

Cross-filtering between visuals ensures that selections in one chart update related visuals in real time.

3.4. Best Practices



3.4.1. Avoid circular relationships

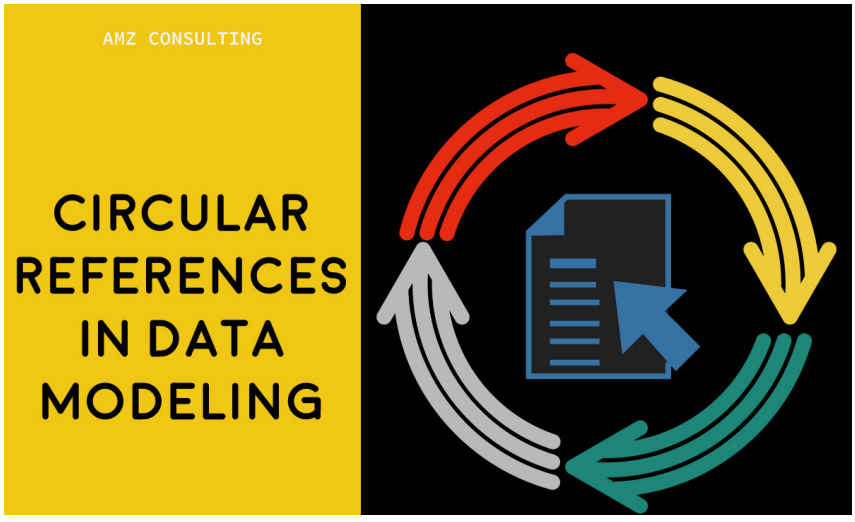
- Circular (many-to-many) relationships complicate the model and can cause incorrect results.
- Prefer a clean one-to-many relationship structure.

3.4.2. Keep data models simple

- Use a star schema whenever possible.
- Remove unnecessary columns or tables to reduce complexity and improve performance.

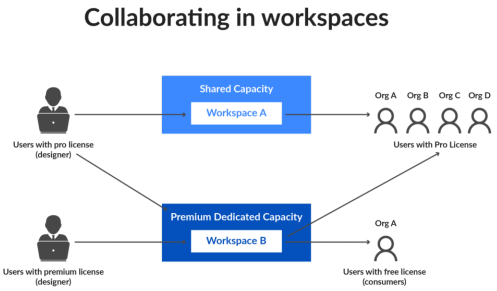
3.4.3. Use descriptive column and table names

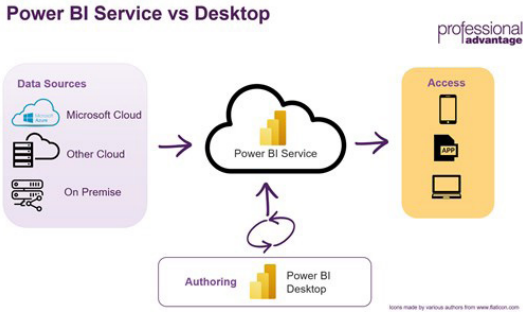
- Rename fields and tables with business-friendly names.
- Example: Instead of tbl_Cust, use Customers.
- Improves readability and makes reports easier for non-technical users to understand.



In short: Data Modeling = structure, DAX = intelligence.
Together, they transform Power BI from a simple charting tool into a true decision-making platform.

Chapter 4: Power BI Service and Collaboration





The Power BI Service (PowerBI.com) is the cloud-based platform where users publish, share, and collaborate on reports and dashboards. While Power BI Desktop is primarily used for creating reports, the Service is designed for distribution, collaboration, and real-time monitoring.

This chapter explains the main features of the Power BI Service and how teams can work together effectively.

4.1. Publishing Reports

- Reports built in Power BI Desktop can be published directly to the Power BI Service.
- Publishing makes reports available online, so users can access them from a browser or mobile app.
- Once published, reports can be organized into workspaces, where teams collaborate and manage content.



4.2. Dashboards

- A dashboard in Power BI Service is a single-page, high-level view that combines visuals from one or more reports or datasets.
- Dashboards allow users to monitor KPIs (Key Performance Indicators) at a glance.
- Tiles on dashboards can be pinned from reports, ensuring that the most important insights are always visible.



4.3. Sharing

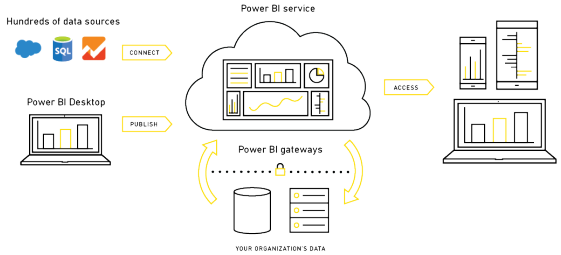
4.3.1. Share dashboards or reports with individuals or groups

- Reports and dashboards can be shared directly with team members or larger groups.
- Sharing ensures everyone works with the same version of the data.
- Content can also be published as an app for wider distribution.

4.3.2. Control permissions (view or edit)

- The owner of a report or dashboard can set access levels:
  - View Only – Recipients can see the content but not modify it.
  - Edit Access – Allows others to update reports, datasets, or dashboards.
- Proper permission control helps maintain security and data integrity.

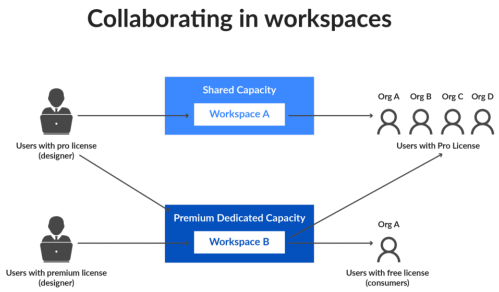
4.4. Data Refresh



- Reports in Power BI Service can be scheduled to refresh automatically, keeping data up to date.
- A gateway may be required for on-premises sources like SQL Server.
- Automatic refresh ensures decision-makers always see the latest insights without manual updates.

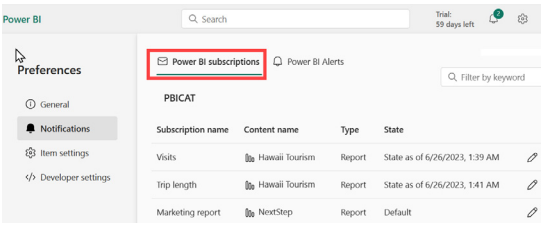
4.5. Collaboration Features

4.5.1. Comments for team feedback



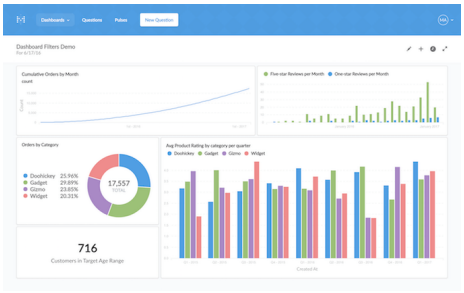
- Users can leave comments on dashboards and reports.
- This allows teams to discuss insights directly within Power BI, reducing the need for separate emails or chats.

4.5.2. Subscriptions to receive report updates via email



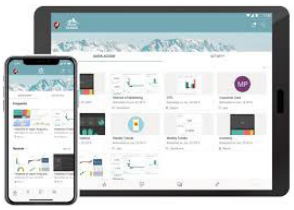
- Users can subscribe to a report or dashboard to receive scheduled email snapshots.
- Subscriptions help keep teams informed without requiring them to log in every day.

4.5.3. Apps to bundle dashboards and reports for distribution



- Power BI allows organizations to create apps that package multiple reports and dashboards.
- Apps make it easier to distribute content at scale, ensuring that users across departments see the same trusted data.

Chapter 5: Advanced Features & Power BI Mobile







As users become more experienced with Power BI, advanced features help unlock deeper insights, improve report security, and enhance usability. In addition, the Power BI Mobile app ensures that reports and dashboards are always available on the go. This chapter explores these advanced capabilities and offers best practices for building effective solutions.

5.1. Power BI Mobile

Power BI Mobile brings business intelligence directly to smartphones and tablets, ensuring access to insights anytime, anywhere.

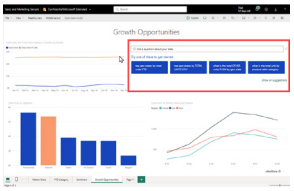
5.1.1. Access dashboards on smartphones and tablets

- Power BI Mobile apps are available for iOS, Android, and Windows devices.
- Users can log in and view the same dashboards and reports published in the Power BI Service.
- Offline caching allows limited access to reports even without an internet connection.

5.1.2. Interact with visuals and drill through reports

- Mobile users can filter, slice, and interact with visuals just like on desktop.
- Reports are optimized for touch, making it easy to explore data by tapping and swiping.
- Drill-through features allow users to move from summary views to detailed insights seamlessly.

5.2. Q&A Featur



- Power BI's Q&A allows users to ask natural language questions about their data.
- Example: Typing "Total sales by region last year" instantly generates a relevant chart or table.
- Q&A makes data exploration accessible for non-technical users and supports conversational analytics.

5.3. Row-Level Security (RLS)

- RLS restricts data access based on a user's role or identity.
- Example: A sales manager can see all regions, but a salesperson only sees their assigned territory.
- RLS ensures sensitive information is protected while still enabling broad report sharing.



5.4. Advanced Visuals



Beyond the built-in charts and tables, Power BI supports custom visuals from the marketplace.

Examples include:

Heatmaps
Funnel charts
KPI cards
Forecasting visuals

Advanced visuals enhance storytelling and provide specialized insights tailored to business needs.

5.5. Best Practices



5.5.1. Design dashboards for clarity and usability

- Keep dashboards focused on key metrics.
- Avoid clutter by limiting the number of visuals per page.

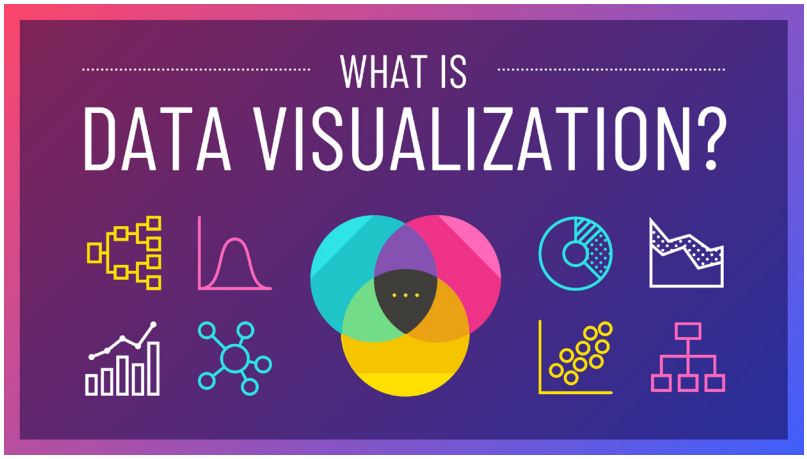
5.5.2. Use concise titles and labels

- Titles should clearly communicate what each chart represents.
- Labels should be short but meaningful to improve readability.

5.5.3. Ensure mobile-friendly layouts for small screens

- Arrange visuals with responsive design in mind.
- Test dashboards on mobile devices to ensure charts and KPIs are easy to view and interact with.

Chapter 6: Data Visualization Best Practices



Effective visualization in Power BI is critical for turning data into actionable insights. Without clear visuals, even the most important data can be ignored or misunderstood. Good visualization allows users to quickly grasp trends, patterns, and key metrics.

Principles of Good Visualization

Creating impactful dashboards starts with foundational principles:

Keep visuals simple and clear

- Avoid unnecessary decorations or complex chart types that do not add value.
- Focus on the story the data is telling, not on flashy effects.
- Example: A simple bar chart showing monthly sales is often more effective than a 3D chart with extra shading.

Avoid unnecessary colors and effects

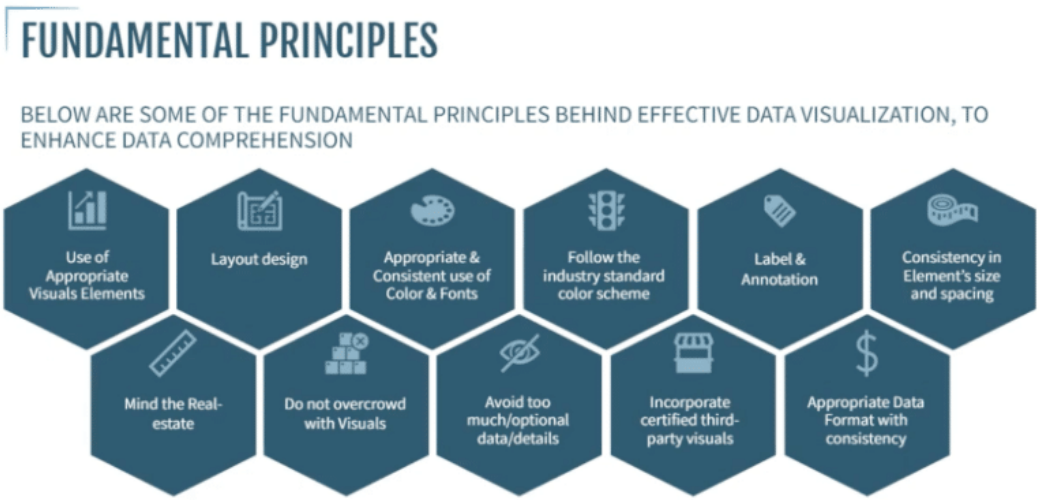
- Use a consistent color palette to maintain clarity.
- Avoid bright, clashing colors or gradients that distract from the data.
- Use color purposefully: e.g., red for negative trends, green for positive performance.

Highlight key metrics and trends

- Use color, size, or placement to draw attention to important values.
- Example: Highlight top-selling products or declining revenue areas.
- Consider using cards or KPI visuals for critical metrics.

Ensure readability with consistent fonts and scales

- Text should be legible at a glance; avoid overly decorative fonts.
- Axis scales should start at zero where appropriate to avoid misleading interpretations.
- Maintain consistent labeling and units across charts.



Common Visuals in Power BI

Power BI provides multiple visual types, each suited for a specific purpose:

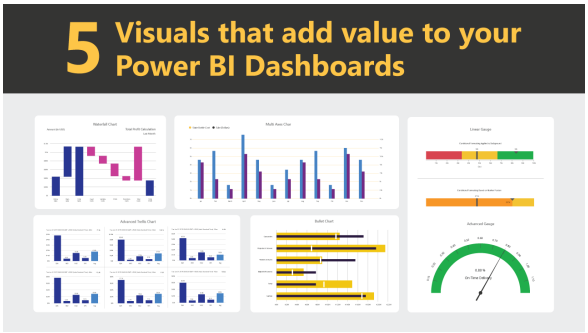
Bar and Column Charts
- Used for comparing categories or time periods.
- Example: Comparing sales by region or product category.

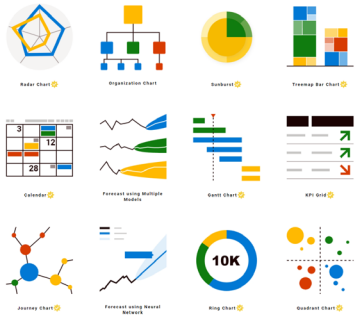
Line Charts
- Ideal for showing trends over time.
- Example: Tracking monthly revenue growth over a year.

Pie and Donut Charts
- Best for showing percentages or parts of a whole.
- Use only when the number of categories is small (3–6) to maintain clarity.

Maps
- Visualize geographic data and regional differences.
- Example: Showing sales distribution by city or country.

Cards
- Display key performance indicators (KPIs) or summary numbers prominently.
- Example: Total revenue, number of active customers, or average order value.





Design Tips for Effective Dashboards

Use filters and slicers
- Allow users to interact with the data by filtering by time, category, or region.
- Example: A sales dashboard can have a slicer for months or product categories.

Arrange visuals logically for storytelling
- Design dashboards to lead the user from overview → trends → details.
- Place the most important KPIs at the top or in prominent locations.

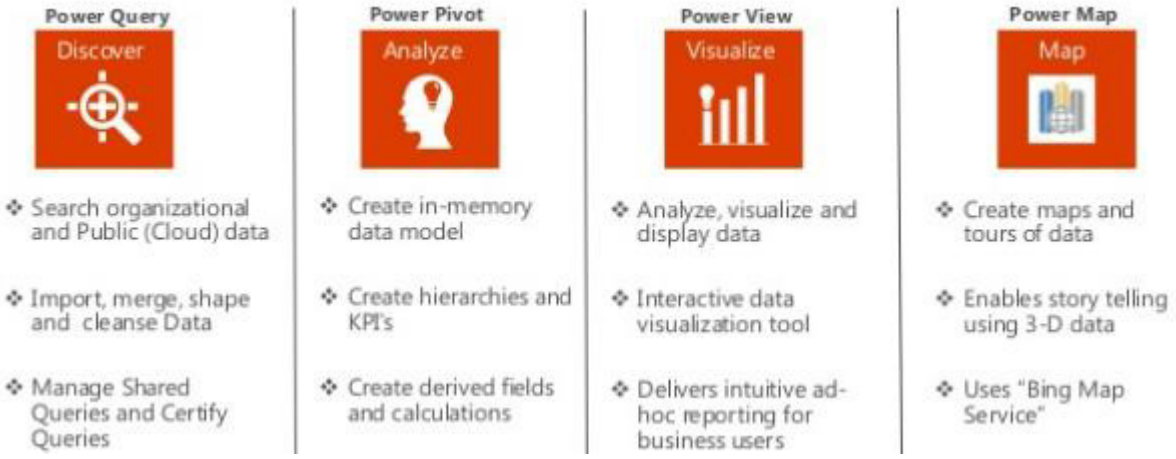
Use color to emphasize, not distract
- Highlight significant trends or deviations using a consistent color scheme.
- Avoid using too many contrasting colors that confuse interpretation.

Include legends, titles, and labels
- Always label axes, charts, and important data points.
- Provide legends for charts with multiple categories.
- Use descriptive titles that clearly state what the chart shows.





Additional Best Practices



Limit the number of visuals per page

- Too many visuals can overwhelm the viewer; fewer, well-chosen charts work better.

Enable drill-downs and hierarchies

- Allow users to explore details without cluttering the dashboard.
- Example: Click on a year to see quarterly and monthly data.

Use tooltips effectively

- Provide additional context when hovering over data points.
- Example: Show product details or revenue breakdowns in a tooltip.

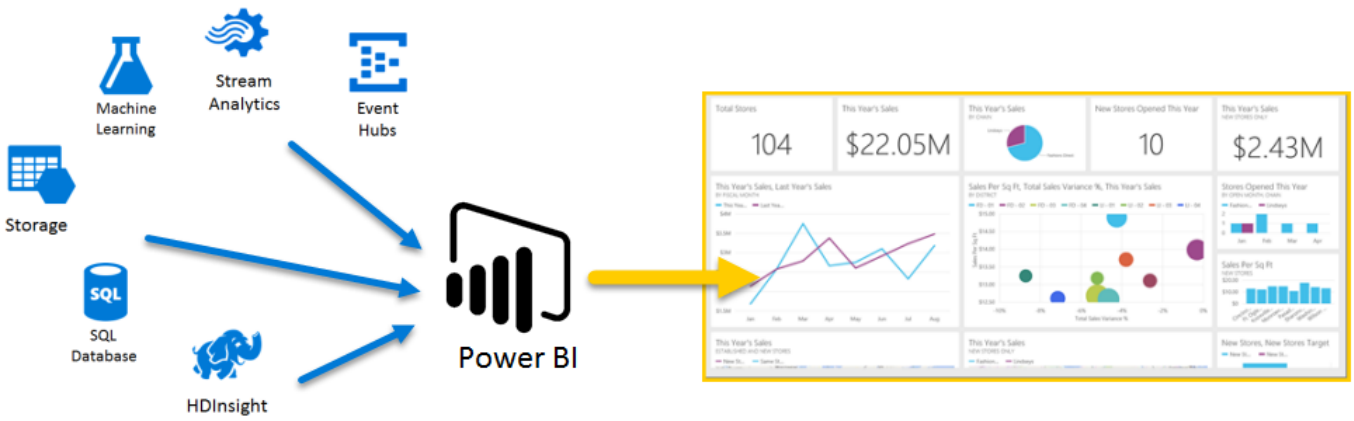
Maintain accessibility

- Ensure color contrast is sufficient for color-blind users.
- Use readable fonts and font sizes.

Conclusion

A successful Power BI dashboard is clear, concise, and interactive. It turns complex datasets into meaningful insights that help users make informed decisions. By following these principles, using the right visuals, and applying good design practices, you can create dashboards that communicate data effectively and drive action.

Chapter 7: Data Connectivity and Integration

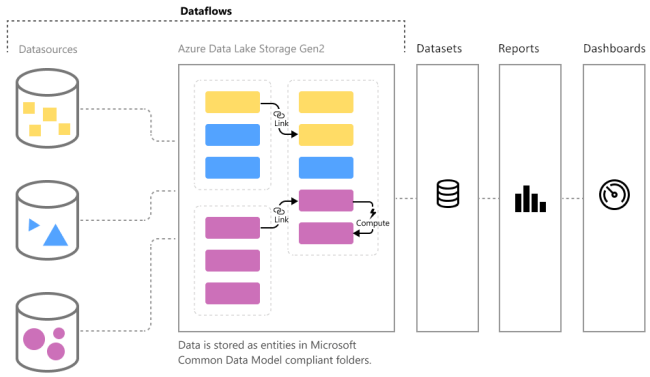


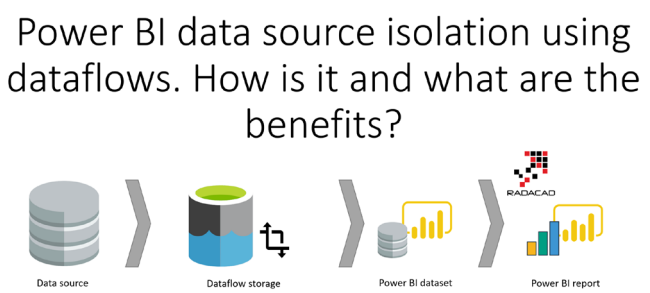
Connecting Power BI to the World of Data

Power BI's strength lies in its ability to connect to a wide variety of data sources, allowing users to bring together information from different platforms into one unified view. Whether your data is stored in local files, databases, or cloud services, Power BI provides flexible and secure connection options for analysis and reporting.

Common Data Connections

Power BI supports a broad range of data sources—both local and cloud-based—making it ideal for diverse business environments.





Power BI supports a broad range of data sources—both local and cloud-based—making it ideal for diverse business environments.

Files

- Excel (.xlsx): One of the most common sources; ideal for importing structured tables and pivot data.
- CSV (Comma-Separated Values): Used for lightweight, flat data files.
- XML / JSON: Great for semi-structured data, often exported from web systems or APIs.

Databases

- SQL Server: Most common enterprise database; fully compatible with both Import and DirectQuery modes.
- Oracle, MySQL, PostgreSQL: Other popular database systems widely used across industries.
- Power BI can connect securely using credentials, enabling direct query execution or data import.

Cloud Services

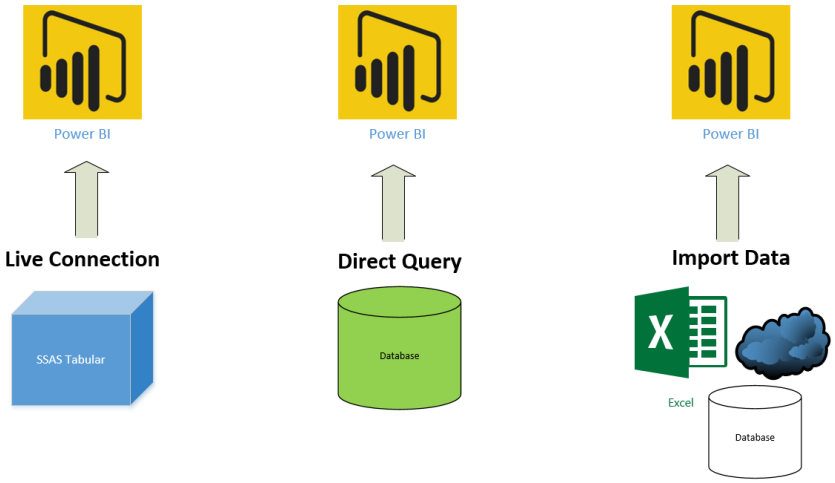
- Microsoft Azure: Connects directly to Azure SQL Database, Azure Blob Storage, and Azure Data Lake.
- SharePoint Online: Import data from lists or document libraries.
- Google Analytics, Salesforce, and other connectors: Access business data hosted on third-party cloud platforms.

APIs and Web Sources

- REST APIs: Retrieve data dynamically from web services.
- Online Data Feeds: Integrate live web data such as financial markets, weather, or social media analytics.

Data Connection Modes

Power BI provides multiple connection modes, allowing flexibility between performance, real-time access, and data freshness.



Import Mode

- Copies data from the source into Power BI's internal storage.
- Offers fast performance and supports full modeling, DAX, and transformations.
- Best for small to medium datasets that don't require real-time updates.

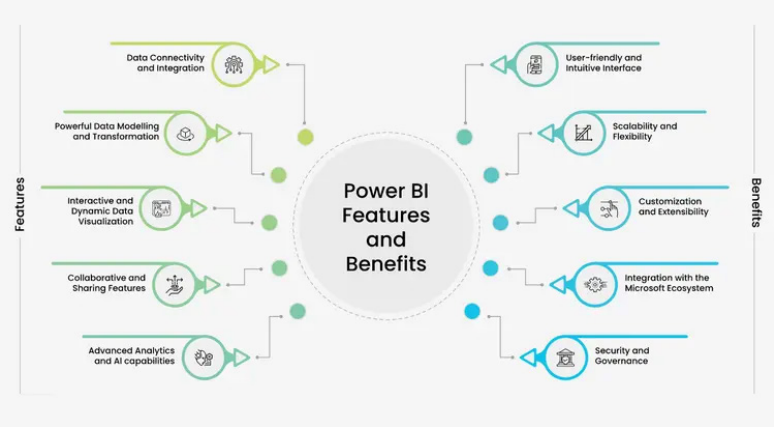
DirectQuery Mode

- Queries data directly from the source every time the report is refreshed.
- Ensures real-time data updates but can be slower depending on source performance.
- Ideal for large datasets or when up-to-date data is critical.

Live Connection

- Connects directly to external data models, such as SQL Server Analysis Services (SSAS) or Azure Analysis Services.
- The model remains managed externally while Power BI acts as the visualization layer.
- Suitable for enterprise environments where data models are centrally maintained.

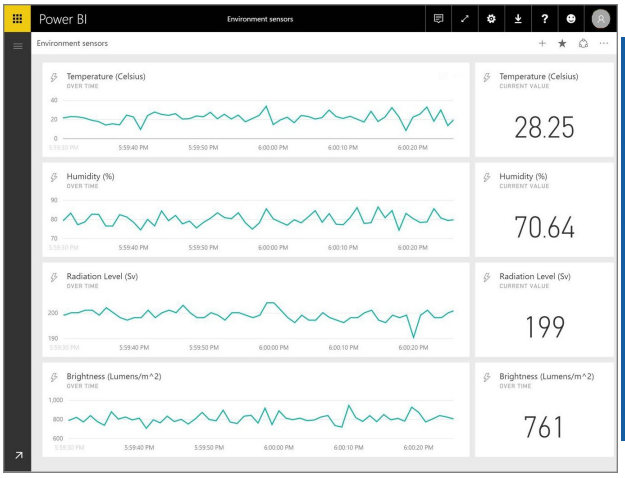
Integration Benefits

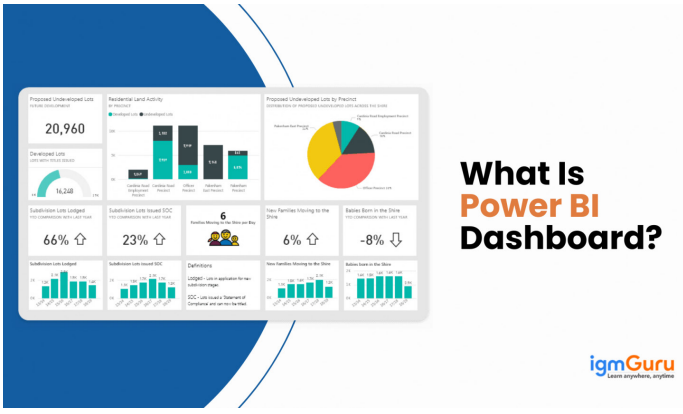


By connecting and integrating multiple data sources, Power BI delivers several powerful benefits:

Real-Time Dashboards

- Stay informed with dashboards that update automatically as new data flows in.
- Example: Monitoring live sales or production data.





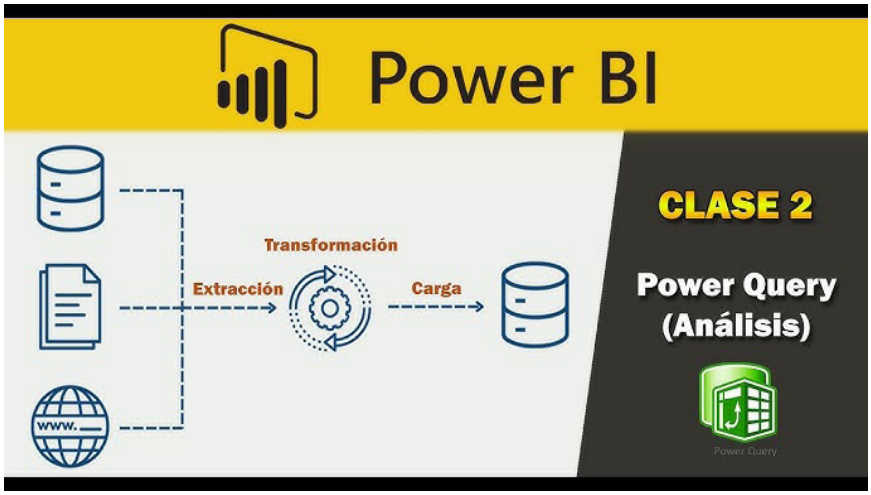
Unified View of Data

- Combine multiple systems—like finance, marketing, and operations—into one cohesive report.
- Helps reveal cross-departmental insights.

Conclusion

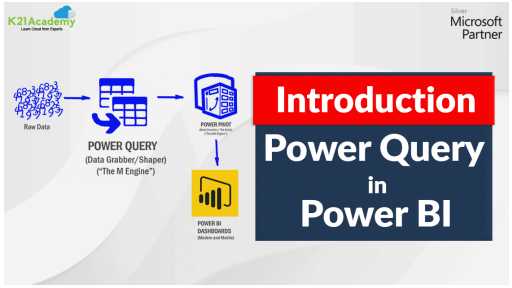
Power BI's robust connectivity and integration capabilities transform it into a central hub for analytics. Whether your data lives in spreadsheets, databases, or the cloud, Power BI brings it all together — empowering users with real-time, unified, and actionable insights.

Chapter 8: Power Query and Data Transformation



Transforming Raw Data into Reliable Insights

Before data can be analyzed in Power BI, it must first be clean, consistent, and well-structured. This is where Power Query comes in — a powerful data preparation tool built into Power BI Desktop. Power Query allows users to connect, clean, transform, and shape data from multiple sources before loading it into reports and dashboards.



What is Power Query?

Power Query is an intuitive tool designed for data extraction, transformation, and loading (ETL).
It helps users:

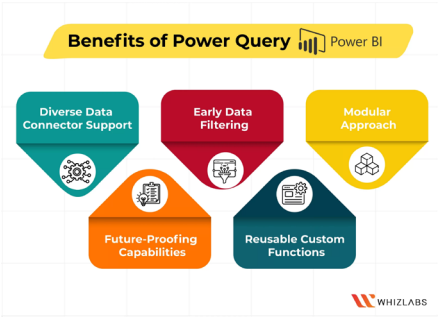
- Clean, filter, and prepare raw data.
- Combine multiple datasets from different sources.
- Apply transformations automatically through step-by-step processes.

All actions in Power Query are recorded as steps that can be refreshed anytime — ensuring data stays up to date and consistent.

Main Features of Power Query

Clean, Filter, and Transform Data

- Remove irrelevant or duplicate records.
- Filter specific rows based on criteria (e.g., only "Completed" transactions).
- Convert messy or inconsistent datasets into structured, usable formats.



Why Power Query Matters

Power Query ensures your data is:

- Clean: Free of duplicates, errors, and inconsistencies.
- Structured: Properly formatted with clear relationships.
- Dynamic: Automatically refreshed when source data changes.



Conclusion

Power Query is the backbone of data preparation in Power BI. By cleaning, transforming, and combining data before analysis, it ensures that every dashboard and report is built on a solid, trustworthy data foundation.